Battery Software Integration Engineer, Energy Storage

Redwood Materials · San Francisco, California, United States

Job ID (this archive)	ENERGY-088
Employer	Redwood Materials
Location(s)	San Francisco, California, United States
Career stage (archive tag)	Experienced (Mid-career)
Location group	Other U.S.
Majors (archive tags)	Mechanical Engineering, Computer Science, Software Engineering
Energy area	Battery storage / software
Pay listed	Not listed
Recruiting status when checked	Open when checked: the employer page had a live application. Checked Oct 26, 2026.
Source page	https://job-boards.greenhouse.io/redwoodmaterials/jobs/6139225004?gh_jid=6139225004
Original link (from the course sheet)	https://boards.greenhouse.io/redwoodmaterials/jobs/6139225004?gh_jid=6139225004
Captured	2026-10-26
Course sheet refs	Expanded archive ENERGY-059

> Saved on 2026-10-26 from the employer's own careers page. The posting text below is the employer's own wording, reformatted for printing. Live listings change and close; check the source page before applying.

Job description

About Redwood Materials

Redwood Materials was founded in 2017 by JB Straubel, co-founder and former CTO of Tesla, on a simple but radical idea: the critical minerals powering the energy transition already exist. They just need to come back. Today Redwood is the largest lithium-ion battery recycler in North America — recovering, refining, and redeploying critical minerals from end-of-life batteries back into the supply chain. On the energy side, Redwood has deployed the largest second-life battery project in the world, delivering power to AI data centers in months rather than years using proprietary energy architecture engineered entirely in-house. The company is profitable, the market it operates in is projected to grow 20 to 50 times over the next decade, and Redwood is the only company with the supply chain, the technology, and the operational scale to lead it.

About the Role

At Redwood, we are architecting the future of grid-scale energy storage and circular energy ecosystems. As we rapidly scale our stationary Battery Energy Storage Systems (BESS), we are looking for a Battery Software Integration Engineer to build the bridge between diverse, second-life automotive batteries and our modern, scalable software platforms.

This role focuses on moving away from rigid, hardware-centric infrastructure toward a resilient, unified, and software-defined energy storage architecture. You will be instrumental in making varied OEM packs operate seamlessly off-vehicle, turning proprietary automotive components into reliable building blocks for the grid. You will serve as the lead Software POC for external OEMs, steering the critical technical discussions necessary to successfully adapt and integrate their battery systems into our architecture

Key Responsibilities

- Technical OEM Partnerships: Be the lead engineering voice bridging the gap between legacy automotive hardware and modern software infrastructure. Dive into the technical weeds with tier-1 suppliers to decode firmware, negotiate engineering solutions, and build the software handshakes that make second-life integration a reality.
- Architectural Integration: Design and deploy translation layers and software gateways to seamlessly integrate vendor-off-the-shelf and second-life OEM battery packs into our centralized energy storage system.

- Modern Software Development: Develop robust, containerized applications to handle external BMS communication, data parsing, and fault management utilizing modern technologies like Rust and Python within Linux environments.
- System Resilience: Build highly resilient communication interfaces (CAN, Ethernet, Modbus) capable of dynamically handling the idiosyncrasies and proprietary logic of different OEM packs without compromising overall system stability.
- Cross-Functional Collaboration: Partner closely with power electronics, site controller, controls teams, and our internal BMS developers to ensure safe, optimized, and harmonized operation of integrated modules within the broader BESS architecture.
- Validation & Testing: Drive rigorous validation across simulation tools, Hardware-in-the-Loop (HIL) platforms, and physical battery hardware to verify OEM BMS behavior, safety boundaries, and control algorithms under novel off-vehicle duty cycles.

Qualifications

- Technical Mastery: Proven track record in controls and software engineering with a deep, practical understanding of lithium-ion battery architectures, external BMS software, and pack-level integration.
- Stakeholder Engagement & POC Experience: Demonstrated experience acting as a primary software POC with external suppliers, automotive OEMs, or internal cross-functional teams. Must possess the communication skills and technical authority to confidently lead complex software discussions, drive alignment across organizational boundaries, and steer reverse-engineering efforts.
- Modern Tech Stack: Proficiency in modern software engineering practices and languages (Rust, Python, C/C++).
- Protocols & Networking: Expertise in automotive and industrial communication networks (CAN, LIN, Automotive Ethernet) and their associated diagnostic toolchains.
- Adaptive Problem Solving: Exceptional capability to reverse-engineer, troubleshoot, and adapt complex OEM electro-mechanical software systems for novel applications in circular energy ecosystems.

In accordance with California pay transparency laws, the salary range for this position is listed below. Actual compensation may vary based on a variety of factors, including experience, education, and skills.

California Pay Range:

$152,000—$287,000 USD

The position is full-time. Compensation will be commensurate with experience.

We collect personal information (PI) from you in connection with your application for employment with Redwood Materials, including the following categories of PI: identifiers, personal records, professional or employment information, and inferences drawn from your PI. We collect your PI for our purposes, including performing services and operations related to your potential employment. If you have additional privacy-related questions, please contact us at privacy@redwoodmaterials.com.

Archive notes for ECS 1100

Retrieval note: New posting found in expansion round. Official posting includes an application form with applicant input fields. Major fit is an archive classification; eligibility follows the employer qualifications.

Degree / experience excerpt: Technical Mastery: Proven track record in controls and software engineering with a deep, practical understanding of lithium-ion battery architectures, external BMS software, and pack-level integration.

Key skills listed in the archive: Python; C++; Rust; Linux

Archive metadata — Accessed October 26, 2026 (America/Chicago).

Recruiting status: Active · PDF: Complete · Source: Official employer Greenhouse Employer / ATS identifier: 6139225004

Career stage: Mid-career

Major fit: Direct skills pathway: software development, computing and engineering; consult the exact degree and experience requirements. Energy-sector connection: Control, integration and infrastructure software for stationary energy storage.

Retrieval note: New posting found in expansion round. Official posting includes an application form with applicant input fields. Major fit is an archive classification; eligibility follows the employer qualifications.

Identity review: New posting: official employer title, location and identifier captured.. Source type: Official employer Greenhouse. Source identifier: 6139225004.

Sources and retrieval details

Employer posting

https://boards.greenhouse.io/redwoodmaterials/jobs/6139225004?gh_jid=6139225004

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https://boards-api.greenhouse.io/v1/boards/redwoodmaterials/jobs?content=true

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